

BEKTAU-ATA, Kazakhstan

WMO: 356990

Lat: 47.450N

Lon: 74.817E

Elev: 620

StdP: 94.10

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.0	-21.8	-28.7	0.3	-22.6	-26.7	0.4	-20.1	15.2	-3.9	12.1	-12.2	3.6	20	0.518

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	33.1	17.7	31.4	17.1	29.9	16.6	18.9	29.8	18.2	28.8	17.6	27.7	2.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.4	11.8	22.1	14.4	11.1	22.1	13.5	10.4	22.1	56.3	29.9	54.1	28.9	52.1	27.7	23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	9.1	7.6	-27.0	35.8	1.9	1.6	-28.3	37.0	-29.4	37.9	-30.5	38.9	-31.9	40.1		
(5)			-27.2	20.8	1.9	1.1	-28.5	21.6	-29.6	22.2	-30.6	22.8	-32.0	23.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.7	-12.1	-10.7	-1.5	9.7	15.8	21.4	22.7	21.6	15.6	7.5	-1.9	-9.2		
	DBStd	13.60	5.13	5.60	6.11	5.61	4.48	3.82	3.66	3.93	5.02	5.25	5.95	5.74		
	HDD10.0	2778	687	581	358	72	7	0	0	11	112	358	594			
	HDD18.3	4682	945	814	614	262	102	16	7	15	111	337	607	852		
	CDD10.0	1556	0	0	2	62	186	341	394	358	178	33	1	0		
	CDD18.3	418	0	0	0	2	23	107	143	115	28	1	0	0		
	CDH23.3	3977	0	0	0	18	193	1003	1381	1122	257	2	0	0		
	CDH26.7	1377	0	0	0	3	36	331	517	418	73	0	0	0		
	WSAvg	2.8	3.4	3.4	3.4	2.7	2.9	2.6	2.6	2.8	2.8	2.5	2.2	2.7		
	PrecAvg	210	18	14	16	16	23	20	28	14	5	13	21	19		
(15) Precipitation	PrecMax	341	34	32	39	35	74	96	82	37	16	36	54	71		
	PrecMin	131	4	0	0	0	3	0	0	0	0	1	1	4		
	PrecStd	47	9	8	9	11	15	19	22	10	5	11	12	12		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.6	4.0	17.1	25.7	29.9	33.2	36.3	35.0	31.8	22.8	14.3	4.2		
		MCWB	-0.9	1.7	8.0	13.0	14.9	18.1	18.8	18.1	15.5	11.5	7.6	0.1		
	2%	DB	-1.2	0.8	13.4	22.7	27.3	31.6	33.4	32.7	28.7	20.0	10.8	1.3		
		MCWB	-2.4	-1.1	6.5	12.0	14.6	17.6	18.1	17.3	14.6	10.1	6.7	-0.8		
	5%	DB	-3.3	-1.0	9.9	20.7	25.4	30.1	31.3	30.8	26.3	17.8	8.0	-0.3		
		MCWB	-4.6	-2.4	5.1	11.1	13.9	17.0	17.4	16.6	13.5	8.9	4.4	-1.9		
	10%	DB	-5.0	-2.8	7.0	18.4	23.7	28.6	29.6	29.0	23.9	15.5	5.8	-1.8		
		MCWB	-6.1	-4.0	3.6	10.1	13.2	16.5	16.9	16.1	12.8	8.0	2.6	-3.2		
	0.4%	WB	-0.5	1.4	9.6	14.2	17.3	19.7	20.0	19.4	16.7	13.1	10.0	1.1		
		MCDB	0.2	3.4	15.0	21.8	26.6	29.7	30.7	31.4	29.8	19.6	12.7	2.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-2.4	-0.7	7.4	12.7	15.8	18.6	19.1	18.3	15.3	11.1	7.2	-0.5		
		MCDB	-1.3	0.6	12.5	21.0	24.5	29.0	29.9	29.9	26.5	18.5	9.8	1.0		
	5%	WB	-4.3	-2.4	5.3	11.9	14.8	17.9	18.4	17.5	14.2	9.8	4.9	-1.8		
		MCDB	-3.4	-1.2	9.1	19.6	23.2	28.2	28.8	28.7	24.9	16.5	7.8	-0.4		
	10%	WB	-6.2	-4.0	3.6	10.7	14.0	17.2	17.8	16.8	13.2	8.4	2.9	-3.2		
		MCDB	-5.1	-2.8	7.0	17.4	22.6	26.9	27.7	27.2	22.8	14.4	5.6	-2.0		
	5% DB	MDBR	6.3	7.1	7.6	10.3	11.1	10.7	10.4	11.0	11.0	9.6	7.3	6.4		
		MCDBR	6.9	8.0	11.0	12.4	12.4	11.8	11.7	12.7	13.4	12.1	9.3	6.6		
		MCWBR	5.9	6.7	6.9	6.3	5.8	4.9	4.6	5.0	5.8	6.4	6.0	5.5		
		MCWBR	6.2	7.8	10.2	11.2	11.6	10.9	10.8	11.6	12.6	11.0	8.7	6.2		
(35) Mean Daily Temperature Range	5% WB	MCWBR	5.6	6.9	6.9	6.0	5.9	4.7	4.5	4.9	6.0	6.5	6.2	5.5		
	taub		0.297	0.327	0.377	0.417	0.422	0.418	0.432	0.386	0.344	0.344	0.328	0.300		
		taud	2.077	2.013	2.059	2.085	2.123	2.204	2.176	2.294	2.371	2.283	2.200	2.122		
(42) Clear-Sky Solar Irradiance	Ebn at Noon		748	806	826	832	842	848	828	855	863	792	712	694		
		Edh at Noon	92	123	139	150	150	139	141	120	101	94	82	79		
	RadAvg		1.70	2.77	4.00	5.42	6.67	6.98	6.59	6.20	4.84	2.96	1.64	1.20		
(45) All-Sky Solar Radiation	RadStd		0.19	0.21	0.30	0.33	0.36	0.40	0.47	0.30	0.26	0.23	0.16	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	+0.52	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page